

Language Recognition system

Advanced Data Sciene & Ai

Team 32

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Project idea & problem

- The Core Problem: Communication barriers faced by hearing-impaired individuals in daily life.
- Project Idea: An intelligent machine learning pipeline designed to recognize and classify hand sign landmarks from extracted datasets (dataset.csv).
- Objective: Automate hand sign translation into actionable data to bridge communication gaps efficiently using AI models.

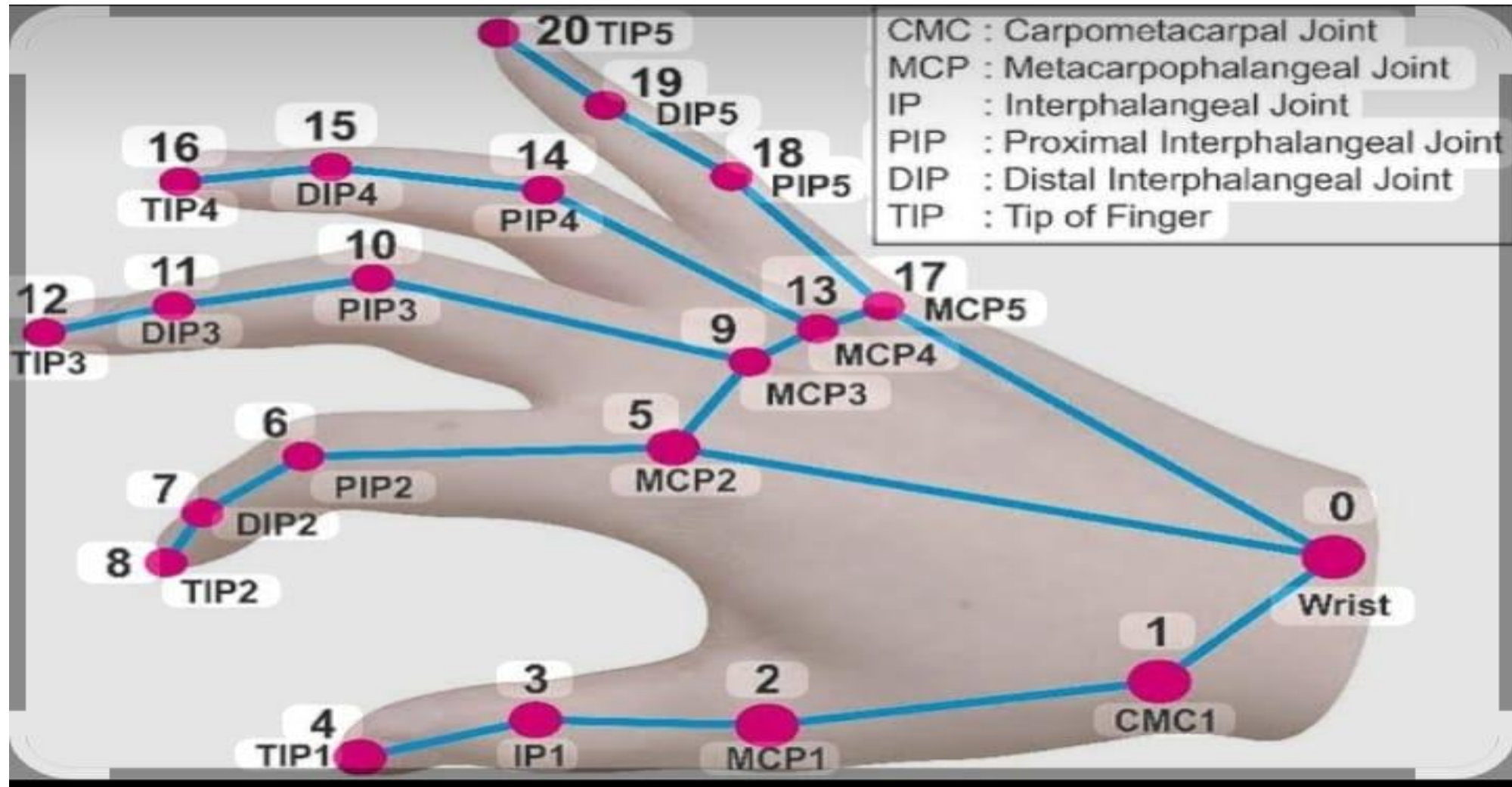
Project pipeline

- Data Collection: Gathering diverse image datasets for each target class/gesture.
- Feature Extraction: Using MediaPipe Hands to isolate 21 2D hand landmarks (x, y).
- Data Preprocessing: Wrist normalization, cleaning, and Tabular CSV generation.
- Model Training & Comparison: Evaluating multiple algorithms (Random Forest, SVM, KNN) using scikit-learn.
- Deployment / Real-Time Inference: Live OpenCV webcam integration utilizing the trained model and standard scaler.

MediaPipe Feature Extraction

- Why MediaPipe? Highly optimized, real-time hand tracking framework that detects hand joint coordinates efficiently.
- Extraction Process:
- ***Images are read using OpenCV and converted from BGR to RGB format (`cv2.cvtColor(img, cv2.COLOR_BGR2RGB)`), because MediaPipe internally expects RGB input arrays.***
- ***Detects 21 distinct hand landmarks (spanning fingers, palm, and wrist).***
- Dimensionality Reduction: Converts high-dimensional raw pixels into compact numerical tabular features (42 coordinate features for x and y per hand).

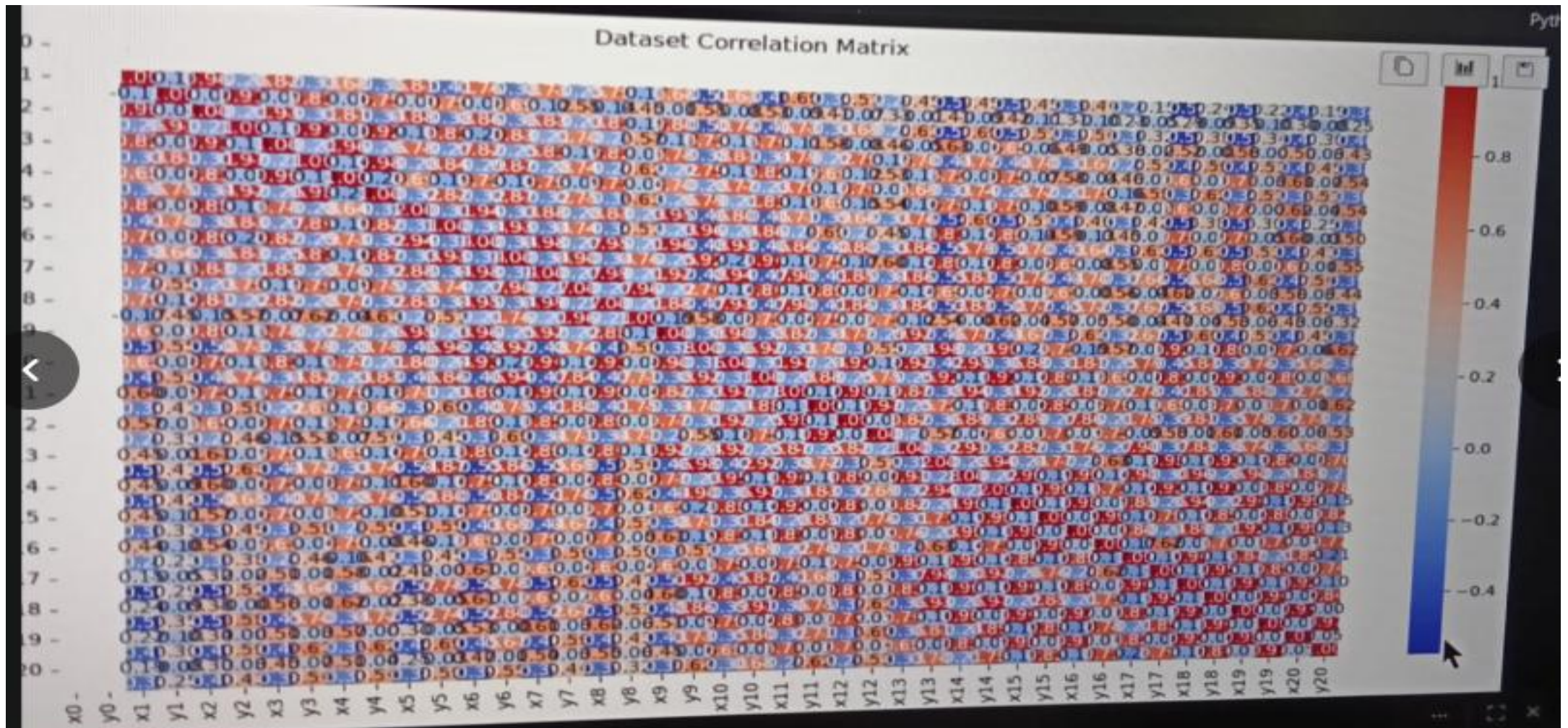
Landmarker 21



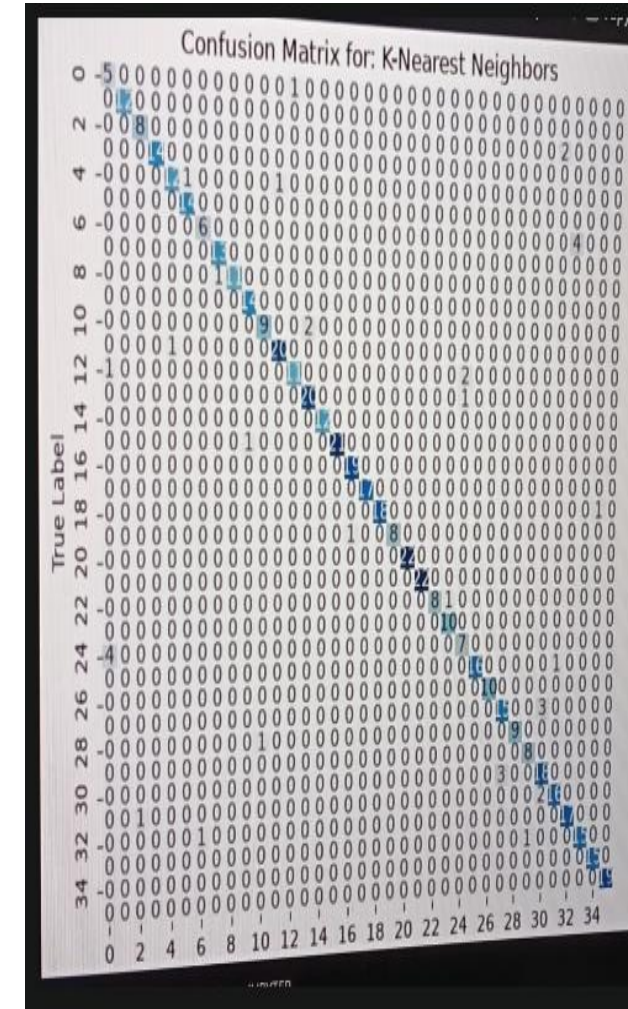
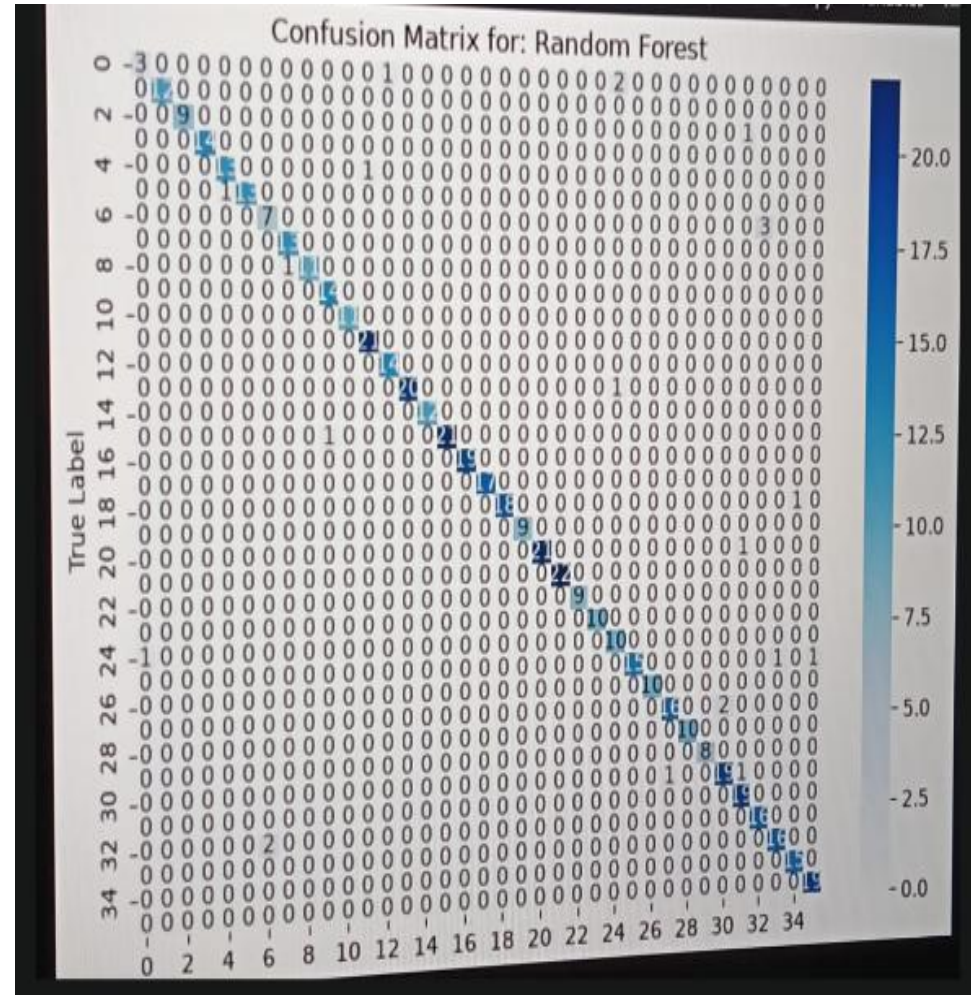
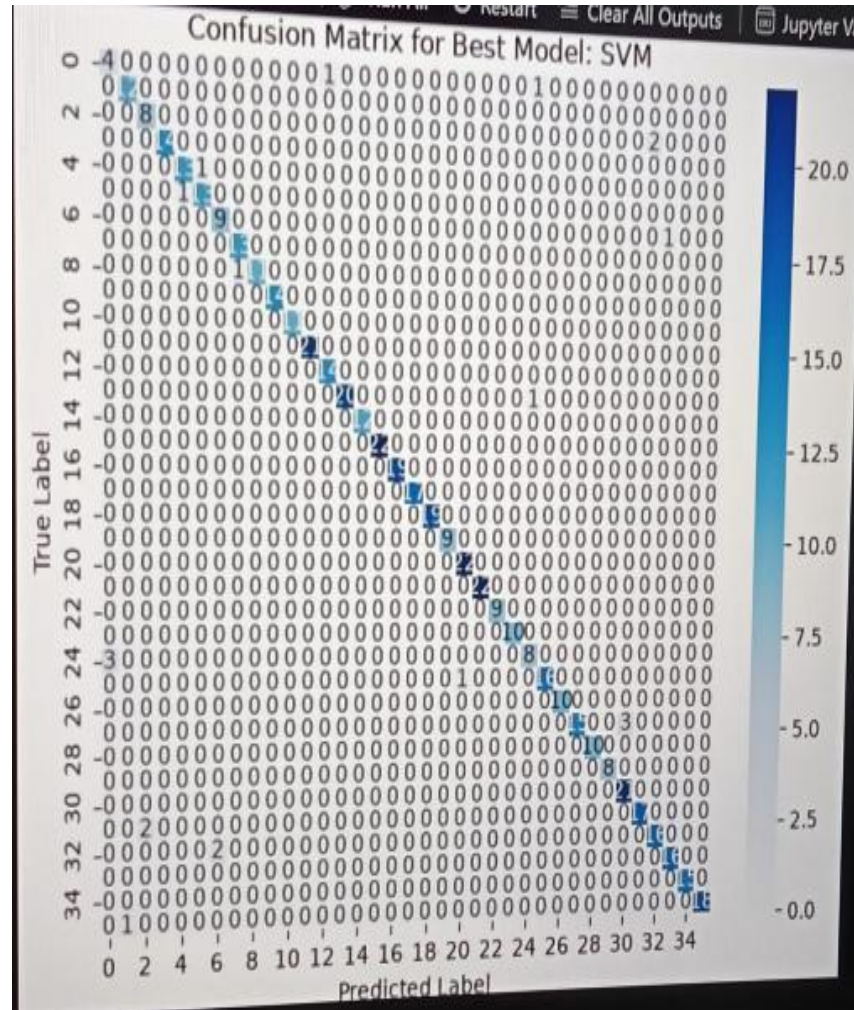
Data Preprocessing & EDA

- Wrist Normalization: Translates all landmark coordinates relative to the wrist landmark (index 0) to ensure invariance to hand position, distance, and scale:
 - $\text{Feature} = X - \text{Wrist}$, $\text{Feature} = Y - \text{Wrist}$.
- Tabular Data Transformation: Compiles processed coordinates and class labels into a structured dataset (hand_gesture_data.csv).
- Scaling: Applied StandardScaler to normalize feature distributions, preventing features with larger numeric ranges from dominating model training.

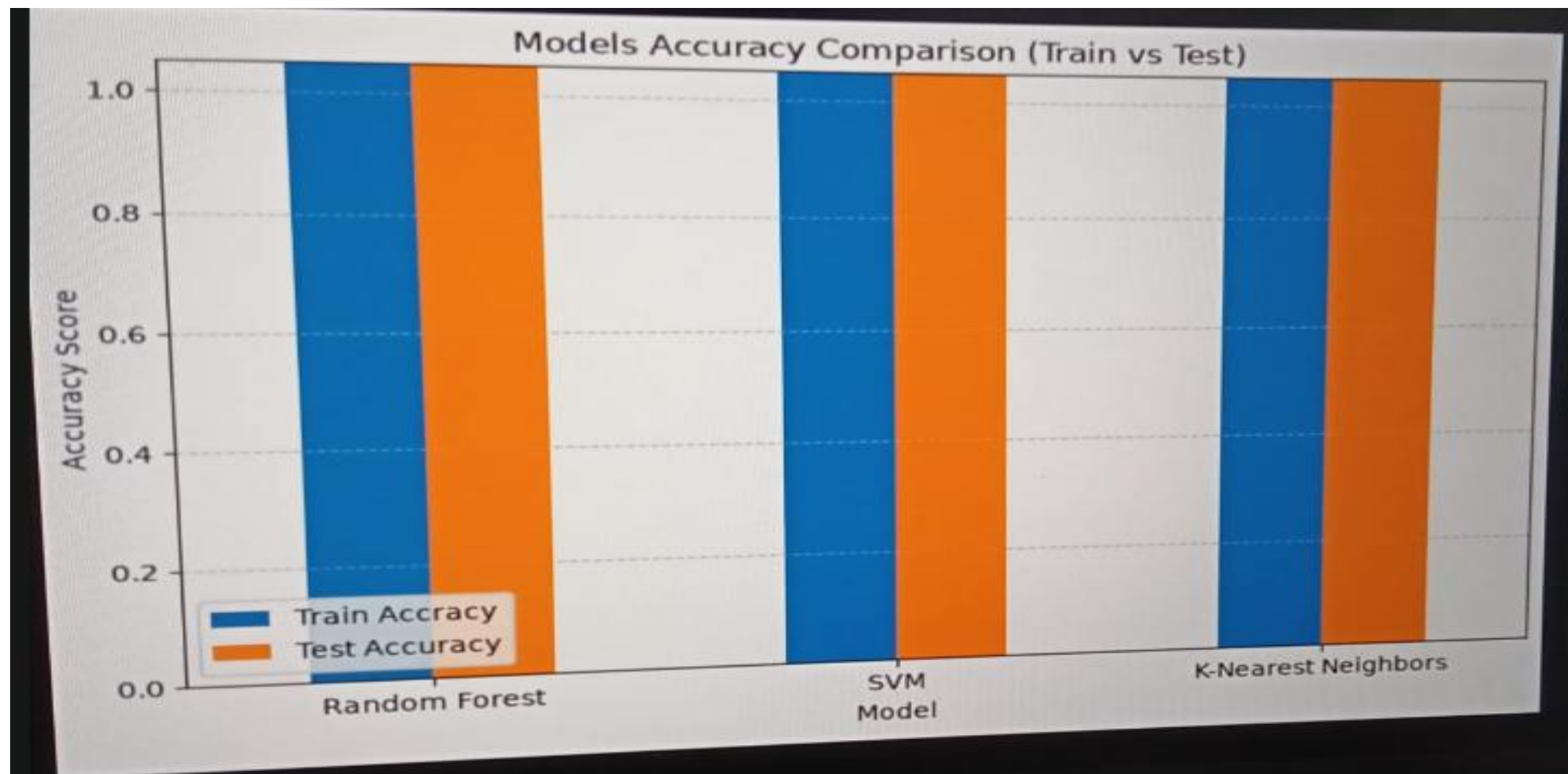
Correlation Matrix



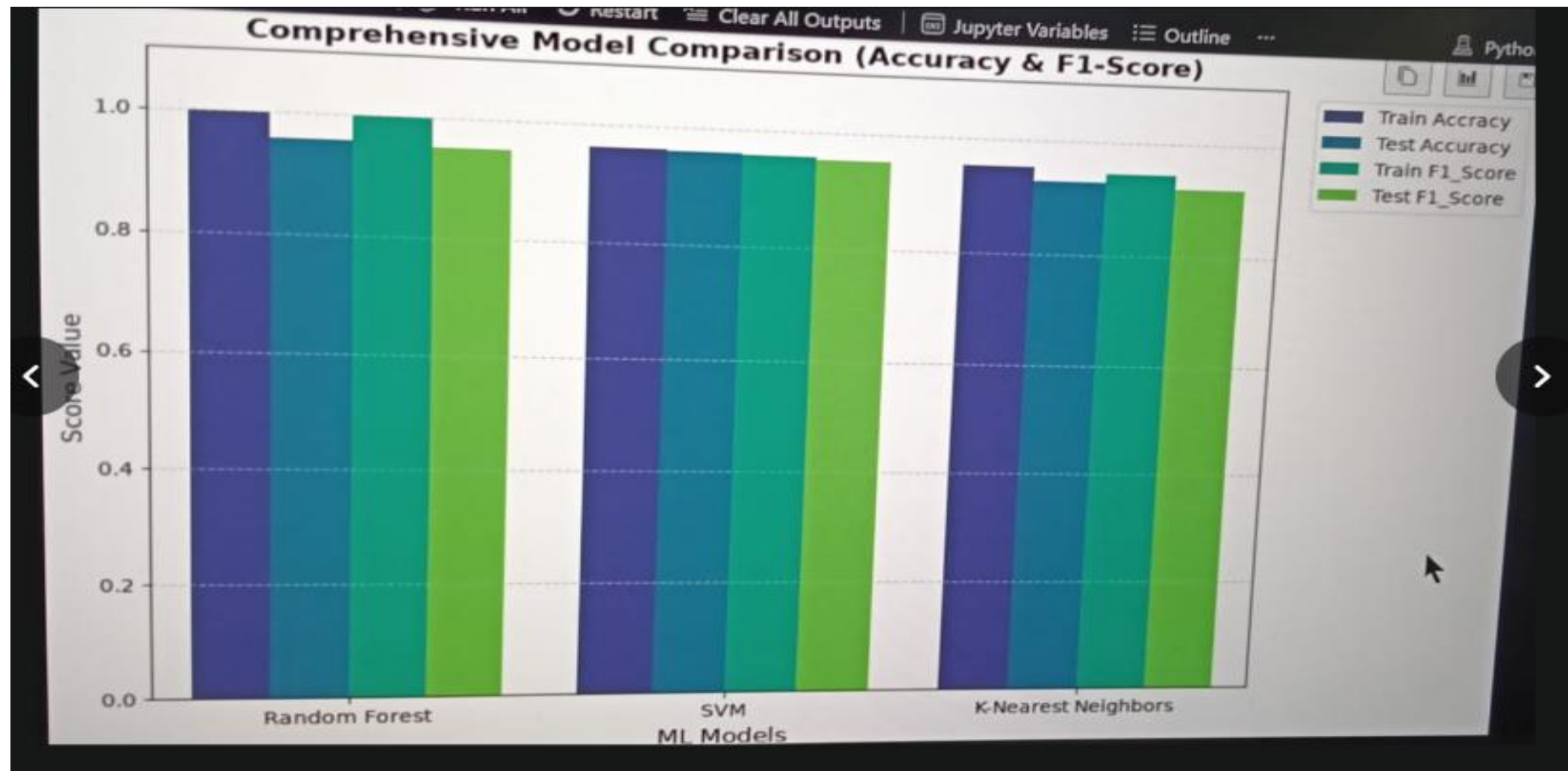
Confusion matrix



Model comparsion for accuracy



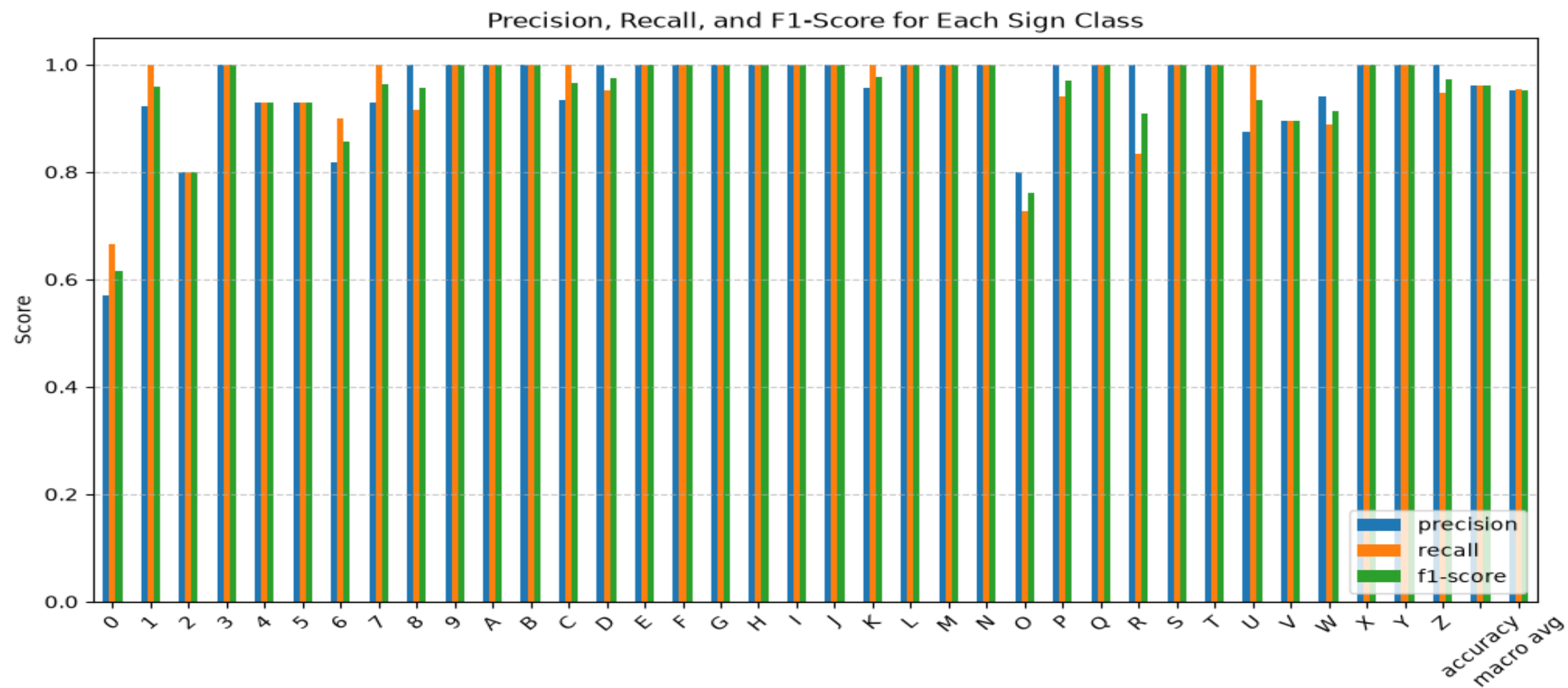
Model comparsion for accuracy& f1-score



Comparision time between models



Metrics



Model Training & Comparison

- Train-Test Split: Data divided into training and testing sets (stratified split to maintain class balance).
- Algorithms Evaluated:
- Random Forest Classifier: Ensemble learning via multiple decision trees.
- Support Vector Machine (SVM): Effective high-margin hyperplane separation using an RBF kernel.
- K-Nearest Neighbors (KNN): Instance-based proximity classification.
- Performance Metric: Evaluated using accuracy score to select the top-performing architecture.

Why SVM Achieved Highest Accuracy

- **Maximized Margin:** Support Vector Machines excel at finding an optimal hyperplane that separates high-dimensional spatial data points cleanly.
- **Non-Linear Mapping:** Utilizing the Radial Basis Function (RBF) kernel allows the model to capture complex, non-linear boundaries between intricate hand gesture shapes.
- **Robustness:** Less prone to overfitting on normalized coordinate vectors compared to simpler instance-based models.

Real-Time Inference Pipeline

- Live Capture: OpenCV opens the webcam stream frame-by-frame.
- Pipeline Execution:
- Capture frame : Convert BGR to RGB for MediaPipe processing.
- Extract 21 hand landmarks and apply wrist normalization.
- Transform feature array using the saved StandardScaler (scaler.transform).
- Predict the gesture class using the optimized SVM model (model.predict).
- Visualization: Renders skeleton connections (mp.hands.HAND_CONNECTIONS) and overlays the predicted sign label onto the live video feed in real-time.

Evaluation Matrices

```
=====Random Forest=====
-> Train Accuracy : 100.00%
-> Test Accuracy :95.65%
[INFO] Generalization is good (No severe overfitting).
```

```
=====SVM=====
-> Train Accuracy : 96.26%
-> Test Accuracy :96.03%
[INFO] Generalization is good (No severe overfitting).
```

```
=====K-Nearest Neighbors=====
-> Train Accuracy : 95.27%
-> Test Accuracy :92.82%
[INFO] Generalization is good (No severe overfitting).
```

MODEL COMPARISON TABLE

Model	Train Accuracy	Test Accuracy	Train F1_Score	Test F1_Score	Time (s)
Random Forest	100.00	95.65	1.0000	0.9500	1.25
SVM	96.26	96.03	0.9570	0.9523	0.07

...

Best Model is SVM

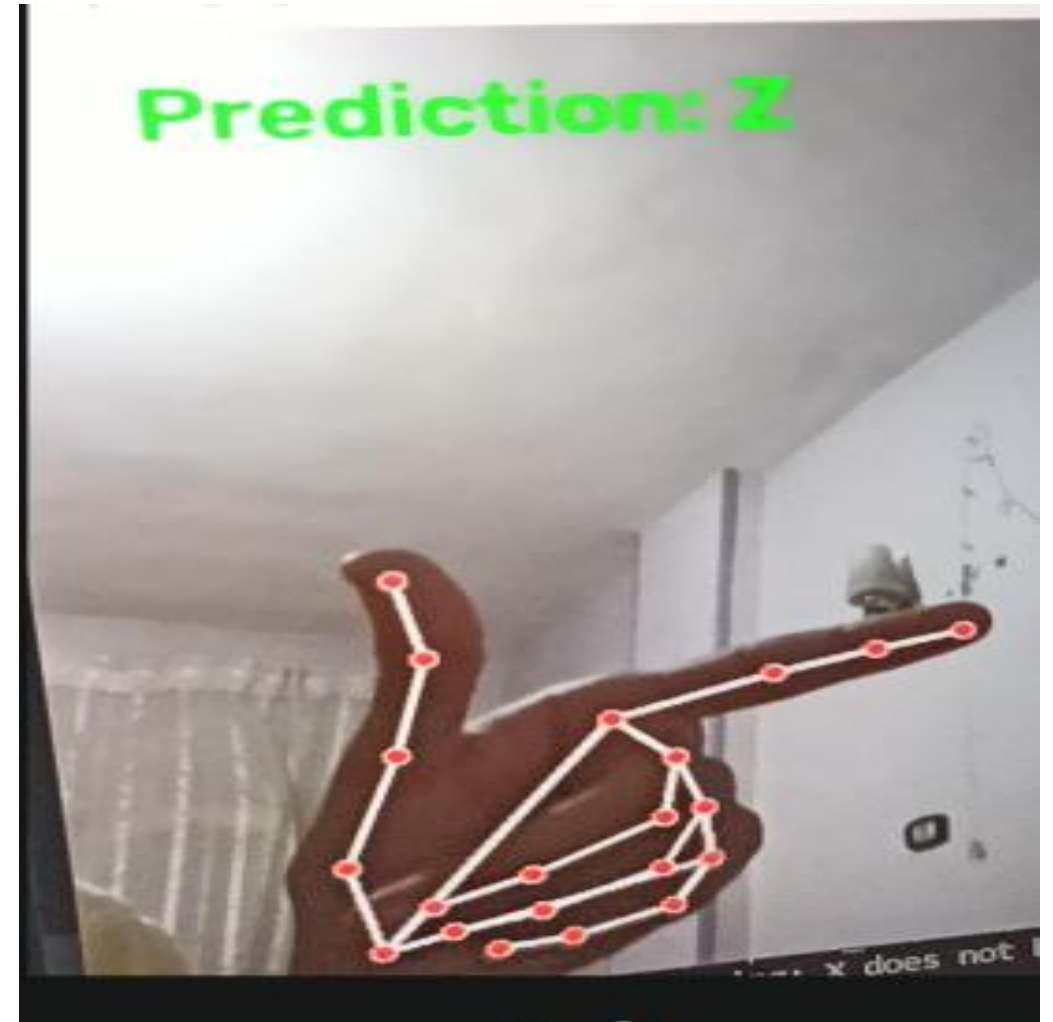
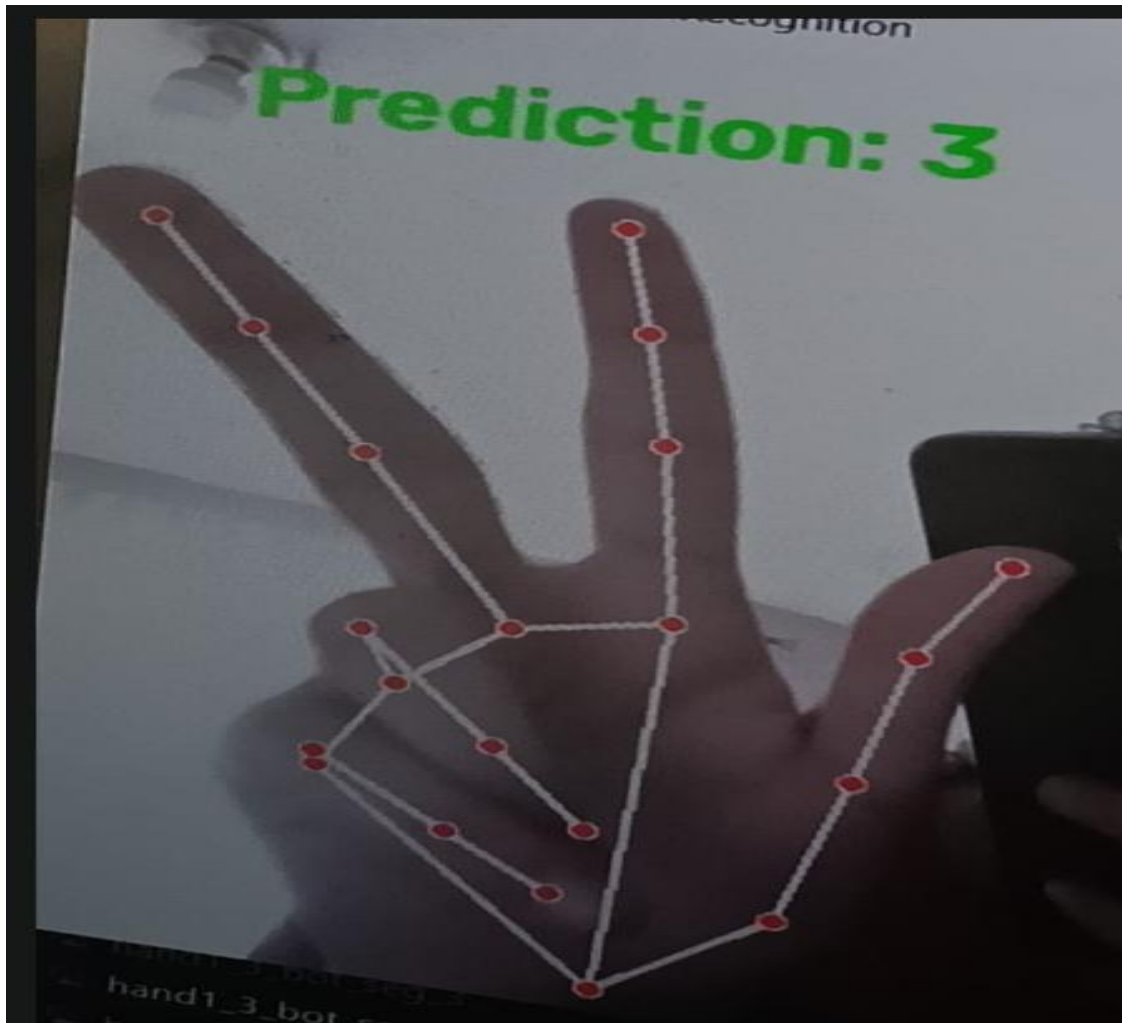
[WINNER] Best Model is 'SVM' with Test Accuracy: 96.03%

[SUCCESS] Best model successfully saved to best_sign_model.pkl

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	Model	Train Accuracy	Test Accuracy	Train F1_Score	Test F1_Score	Time (s)
0	Random Forest	100.00%	95.65%	1.0000	0.9500	1.47s
1	SVM	96.26%	96.03%	0.9570	0.9523	0.09s
2	K-Nearest Neighbors	95.27%	92.82%	0.9449	0.9190	0.01s

Detection&output



Conclusion & Future Scope

- Summary: Successfully built a robust, streamlined sign language recognition tool translating raw video streams into accurate text classifications.
- Key Achievements: Automated feature extraction, robust coordinate normalization, rigorous model benchmarking, and smooth real-time performance.
- Future Enhancements: Expanding the dataset to support dynamic continuous signing and deployment to mobile platforms.